

Chapter Summary and Key Terms

CHEMICAL EQUATIONS (INTRODUCTION AND SECTION 3.1)

The study of the quantitative relationships between chemical formulas and chemical equations is known as **stoichiometry**. One of the important concepts of stoichiometry is the **law of conservation of mass**, which states that the total mass of the products of a chemical reaction is the same as the total mass of the reactants. The same numbers of atoms of each type are present before and after a chemical reaction. A balanced **chemical equation** shows equal numbers of atoms of each element on each side of the equation. Equations are balanced by placing coefficients in front of the chemical formulas for the **reactants** and **products** of a reaction, *not* by changing the subscripts in chemical formulas.

SIMPLE PATTERNS OF CHEMICAL REACTIVITY (SECTION 3.2)

Among the reaction types described in this chapter are (1) **combination reactions**, in which two reactants combine to form one product; (2) **decomposition reactions**, in which a single reactant forms two or more products; and (3) **combustion reactions** in oxygen, in which a substance, typically a hydrocarbon, reacts rapidly with O_2 to form CO_2 and H_2O .

FORMULA WEIGHTS (SECTION 3.3) Much quantitative information can be determined from chemical formulas and balanced chemical equations by using atomic weights. The **formula weight** of a compound equals the sum of the atomic weights of the atoms in its formula. If the formula is a molecular formula, the formula weight is also called the **molecular weight**. Atomic weights and formula weights can be used to determine the **elemental composition** of a compound.

AVOGADRO'S NUMBER AND THE MOLE (SECTION 3.4) A mole of any substance contains **Avogadro's number** (6.02×10^{23}) of formula

units of that substance. The mass of a **mole** of atoms, molecules, or ions (the **molar mass**) equals the formula weight of that material expressed in grams. The mass of 1 molecule of H_2O , for example, is 18.0 u, so the mass of 1 mol of H_2O is 18.0 g. That is, the molar mass of H_2O is 18.0 g/mol.

EMPIRICAL FORMULAS FROM ANALYSIS (SECTION 3.5) The empirical formula of any substance can be determined from its percent composition by calculating the relative number of moles of each atom in 100 g of the substance. If the substance is molecular in nature, its molecular formula can be determined from the empirical formula if the molecular weight is also known. Combustion analysis is a special technique for determining the empirical formulas of compounds containing only carbon, hydrogen, and/or oxygen.

QUANTITATIVE INFORMATION FROM BALANCED EQUATIONS AND LIMITING REACTANTS (SECTIONS 3.6 AND 3.7)

The mole concept can be used to calculate the relative quantities of reactants and products in chemical reactions. The coefficients in a balanced equation give the relative numbers of moles of the reactants and products. To calculate the number of grams of a product from the number of grams of a reactant, first convert grams of reactant to moles of reactant. Then use the coefficients in the balanced equation to convert the number of moles of reactant to moles of product. Finally, convert moles of product to grams of product.

A **limiting reactant** is completely consumed in a reaction. When it is used up, the reaction stops, thus limiting the quantities of products formed. The **theoretical yield** of a reaction is the quantity of product calculated to form when all of the limiting reactant reacts. The actual yield of a reaction is always less than the theoretical yield. The **percent yield** compares the actual and theoretical yields.

Learning Outcomes After studying this chapter, you should be able to:

- Balance chemical equations. (Section 3.1)
Related Exercises: 3.5, 3.70
- Predict the products of simple combination, decomposition, and combustion reactions. (Section 3.2) *Related Exercises: 3.12, 3.74*
- Calculate formula weights. (Section 3.3) *Related Exercises: 3.17, 3.76*
- Convert grams to moles and vice versa using molar masses. (Section 3.4) *Related Exercises: 3.25, 3.81*
- Convert number of molecules to moles and vice versa using Avogadro's number. (Section 3.4) *Related Exercises: 3.27, 3.83*
- Calculate the empirical and molecular formulas of a compound from percentage composition and molecular weight. (Section 3.5) *Related Exercises: 3.37, 3.90*
- Identify limiting reactants and calculate amounts, in grams or moles, of reactants consumed and products formed for a reaction. (Section 3.6) *Related Exercises: 3.47, 3.96*
- Calculate the percent yield of a reaction. (Section 3.7)
Related Exercises: 3.59, 3.105

Key Equations

$$\% \text{ mass composition of element} = \frac{\left(\frac{\text{number of atoms of element}}{\text{formula weight of compound}} \right) \left(\frac{\text{atomic weight of element}}{\text{of element}} \right)}{\text{formula weight of compound}} \times 100\% \quad [3.15]$$

This is the formula to calculate the mass percentage of each element in a compound. The sum of all the percentages of all the elements in a compound should add up to 100%.

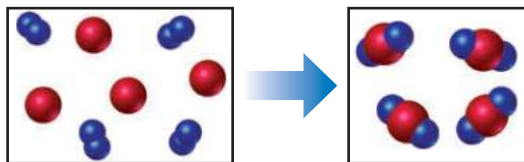
$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

This is the formula to calculate the percent yield of a reaction. The percent yield can never be more than 100%.

Exercises

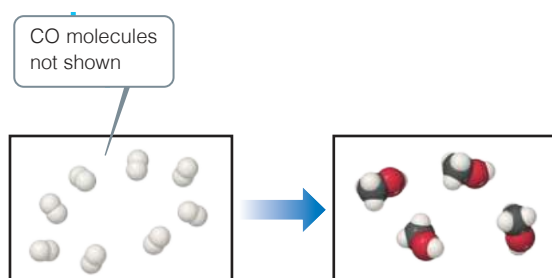
Visualizing Concepts

- 3.61** The reaction between reactant A (blue spheres) and reactant B (red spheres) is shown in the following diagram:

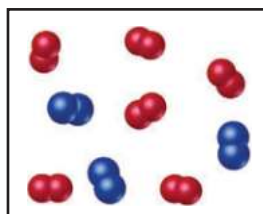


Based on this diagram, which equation best describes the reaction? [Section 3.1]

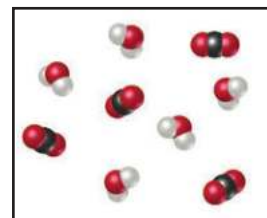
- (a) $A_2 + B \longrightarrow A_2B$
 (b) $A_2 + 4 B \longrightarrow 2 AB_2$
 (c) $2 A + B_4 \longrightarrow 2 AB_2$
 (d) $A + B_2 \longrightarrow AB_2$
- 3.62** The following diagram shows the combination reaction between hydrogen, H_2 , and carbon monoxide, CO , to produce methanol, CH_3OH (white spheres are H, black spheres are C, red spheres are O). The correct number of CO molecules involved in this reaction is not shown. [Section 3.1]
- (a) Determine the number of CO molecules that should be shown in the left (reactants) box.
 (b) Write a balanced chemical equation for the reaction.



- 3.63** The following diagram represents the collection of elements formed by a decomposition reaction. (a) If the blue spheres represent N atoms and the red ones represent O atoms, what was the empirical formula of the original compound? (b) Could you draw a diagram representing the molecules of the compound that had been decomposed? Why or why not? [Section 3.2]



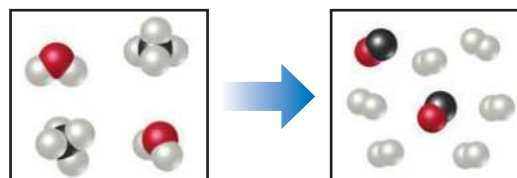
- 3.64** The following diagram represents the collection of CO_2 and H_2O molecules formed by complete combustion of a hydrocarbon. What is the empirical formula of the hydrocarbon? [Section 3.2]



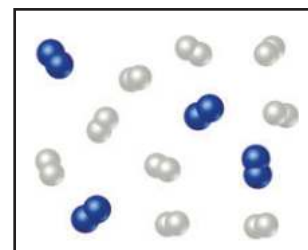
- 3.65** Glycine, an amino acid used by organisms to make proteins, is represented by the following molecular model.
- (a) Write its molecular formula.
 (b) Determine its molar mass.
 (c) Calculate how many moles of glycine are in a 100.0-g sample of glycine.
 (d) Calculate the percent nitrogen by mass in glycine. [Sections 3.3 and 3.5]



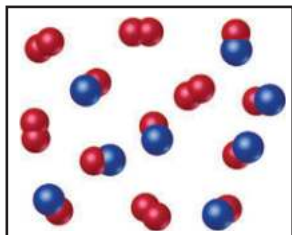
- 3.66** The following diagram represents a high-temperature reaction between CH_4 and H_2O . Based on this reaction, find how many moles of each product can be obtained starting with 4.0 mol CH_4 . [Section 3.6]



- 3.67** Nitrogen (N_2) and hydrogen (H_2) react to form ammonia (NH_3). Consider the mixture of N_2 and H_2 shown in the accompanying diagram. The blue spheres represent N, and the white ones represent H. (a) Write the balanced chemical equation for the reaction. (b) What is the limiting reactant? (c) How many molecules of ammonia can be made, assuming the reaction goes to completion, based on the diagram? (d) Are any reactant molecules left over, based on the diagram? If so, how many of which type are left over? [Section 3.7]



- 3.68** Nitrogen monoxide and oxygen react to form nitrogen dioxide. Consider the mixture of NO and O₂ shown in the accompanying diagram. The blue spheres represent N, and the red ones represent O. **(a)** How many molecules of NO₂ can be formed, assuming the reaction goes to completion? **(b)** What is the limiting reactant? **(c)** If the actual yield of the reaction was 75% instead of 100%, how many molecules of each kind would be present after the reaction was over?



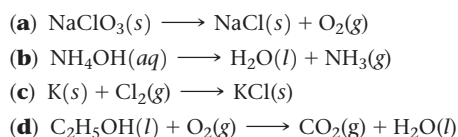
Chemical Equations and Simple Patterns of Chemical Reactivity (Section 3.1)

- 3.69** A key step in balancing chemical equations is correctly identifying the formulas of the reactants and products. For example, consider the reaction between calcium oxide, CaO(s), and H₂O(l) to form aqueous calcium hydroxide. **(a)** Write a balanced chemical equation for this combination reaction, having correctly identified the product as Ca(OH)₂(aq). **(b)** Is it possible to balance the equation if you incorrectly identify the product as CaOH(aq), and if so, what is the equation?
- 3.70** Balance the following equations:
- $\text{HClO}_4(\text{aq}) + \text{P}_4\text{O}_{10}(\text{s}) \longrightarrow \text{HPO}_3(\text{aq}) + \text{Cl}_2\text{O}_7(\text{l})$
 - $\text{Au}_2\text{S}_3(\text{s}) + \text{H}_2(\text{g}) \longrightarrow \text{Au}(\text{s}) + \text{H}_2\text{S}(\text{g})$
 - $\text{Ba}_3\text{N}_2(\text{s}) + \text{H}_2\text{O}(\text{aq}) \longrightarrow \text{Ba}(\text{OH})_2(\text{aq}) + \text{NH}_3(\text{g})$
 - $\text{Na}_2\text{CO}_3(\text{aq}) + \text{HCl}(\text{aq}) \longrightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
- 3.71** Balance the following equations:
- $\text{CF}_4(\text{l}) + \text{Br}_2(\text{g}) \longrightarrow \text{CBr}_4(\text{l}) + \text{F}_2(\text{g})$
 - $\text{Cu}(\text{s}) + \text{HNO}_3(\text{aq}) \longrightarrow \text{Cu}(\text{NO}_3)_2(\text{aq}) + \text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 - $\text{MnO}_2(\text{s}) + \text{HCl}(\text{aq}) \longrightarrow \text{MnCl}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) + \text{Cl}_2(\text{g})$
 - $\text{KOH}(\text{aq}) + \text{H}_3\text{PO}_4(\text{aq}) \longrightarrow \text{K}_3\text{PO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- 3.72** Write balanced chemical equations to correspond to each of the following descriptions: **(a)** When sulfur trioxide gas reacts with water, a solution of sulfuric acid forms. **(b)** Boron sulfide, B₂S₃(s), reacts violently with water to form dissolved boric acid, H₃BO₃, and hydrogen sulfide gas. **(c)** Phosphine, PH₃(g), combusts in oxygen gas to form water vapor and solid tetraphosphorus decoxide. **(d)** When solid mercury(II) nitrate is heated, it decomposes to form solid mercury(II) oxide, gaseous nitrogen dioxide, and oxygen. **(e)** Copper metal reacts with hot concentrated sulfuric acid solution to form aqueous copper(II) sulfate, sulfur dioxide gas, and water.

Patterns of Chemical Reactivity (Section 3.2)

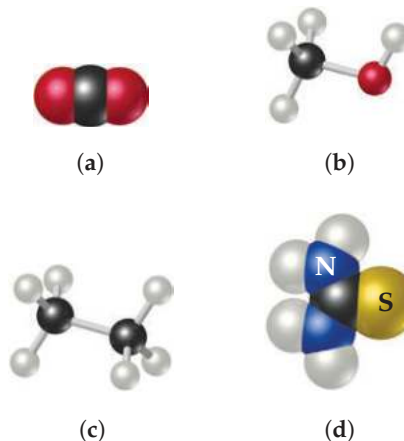
- 3.73** **(a)** When a compound containing C, H, and O is completely combusted in air, what reactant besides the hydrocarbon is involved in the reaction? **(b)** What products form in this reaction? **(c)** What is the sum of the coefficients in the balanced chemical equation for the combustion of one mole of acetone, C₃H₆O(l), in air?
- 3.74** Write a balanced chemical equation for the reaction that occurs when **(a)** titanium metal reacts with O₂(g); **(b)** silver(I) oxide decomposes into silver metal and oxygen gas when heated; **(c)** propanol, C₃H₇OH(l) burns in air; **(d)** methyl *tert*-butyl ether, C₅H₁₂O(l), burns in air.

- 3.75** Balance the following equations and indicate whether they are combination, decomposition, or combustion reactions:



Formula Weights (Section 3.3)

- 3.76** Determine the formula weights of each of the following compounds: **(a)** Butyric acid, CH₃CH₂CH₂COOH, which is responsible for the rotten smell of spoiled food; **(b)** sodium perborate, NaBO₃, a substance used as bleach; **(c)** calcium carbonate, CaCO₃, a substance found in marble. **(c)** CF₂Cl₂, a refrigerant known as Freon; **(d)** NaHCO₃, known as baking soda and used in bread and pastry baking; **(e)** iron pyrite, FeS₂, which has a golden appearance and is known as “Fool’s Gold.”
- 3.77** Calculate the percentage by mass of the indicated element in the following compounds: **(a)** hydrogen in methane, CH₄, the major hydrocarbon in natural gas; **(b)** oxygen in vitamin E, C₂₉H₅₀O₂; **(c)** sulphur in magnesium sulphate, MgSO₄, a substance used as a drying agent; **(d)** nitrogen in epinephrine, C₉H₁₃NO₃, also known as adrenalin, a hormone that is important for the fight-or-flight response; **(e)** oxygen in the insect pheromone sulcatol, C₈H₁₆O; **(f)** carbon in sucrose, C₁₂H₂₂O₁₁, the compound that is responsible for the sweet taste of table sugar.
- 3.78** Calculate the percentage of carbon by mass in each of the compounds represented by the following models:



Avogadro's Number and the Mole (Section 3.4)

- 3.79** **(a)** What is the mass, in grams, of one mole of ⁷⁹Br? **(b)** How many bromine atoms are present in one mole of ⁷⁹Br?
- 3.80** Without doing any detailed calculations (but using a periodic table to give atomic weights), rank the following samples in order of increasing numbers of atoms: 0.2 mol PCl₅ molecules, 80 g Fe₂O₃, 3.0 × 10²³ CO molecules.
- 3.81** Calculate the following quantities:
- mass, in grams, of 1.50 × 10⁻² mol CdS
 - number of moles of NH₄Cl in 86.6 g of this substance
 - number of molecules in 8.447 × 10⁻² mol C₆H₆
 - number of O atoms in 6.25 × 10⁻³ mol Al(NO₃)₃
- 3.82** **(a)** What is the mass, in grams, of 1.223 mol of iron(III) sulfate? **(b)** How many moles of ammonium ions are in 6.955 g of ammonium carbonate?

- (c) What is the mass, in grams, of 1.50×10^{21} molecules of aspirin, $C_9H_8O_4$?
- (d) What is the molar mass of diazepam (Valium®) if 0.05570 mol has a mass of 15.86 g?
- 3.83** The molecular formula of salicylic acid, a compound commonly found in facial cleanser, is $C_7H_6O_3$. (a) What is the molar mass of salicylic acid? (b) How many moles of salicylic acid are present in 0.5 mg of this substance? (c) How many molecules of salicylic acid are in 0.5 mg of this substance? (d) How many oxygen atoms are present in 0.5 mg of salicylic acid?
- 3.84** A sample of the male sex hormone testosterone, $C_{19}H_{28}O_2$, contains 3.88×10^{21} hydrogen atoms. (a) How many atoms of carbon does it contain? (b) How many molecules of testosterone does it contain? (c) How many moles of testosterone does it contain? (d) What is the mass of this sample in grams?
- 3.85** At least 25 μg of tetrahydrocannabinol (THC), the active ingredient in marijuana, is required to produce intoxication. The molecular formula of THC is $C_{21}H_{30}O_2$. How many moles of THC does this 25 μg represent? How many molecules?

Empirical Formulas from Analyses (Section 3.5)

- 3.86** Determine the empirical formula of each of the following compounds if a sample contains (a) 3.92 mol C, 5.99 mol H, and 2.94 mol O; (b) 12.0 g calcium and 2.8 g nitrogen; (c) 89.14% Au and 10.86% O by mass.
- 3.87** Determine the empirical formulas of the compounds with the following compositions by mass:
- (a) 42.1% Na, 18.9% P, and 39.0% O
- (b) 18.7% Li, 16.3% C, and 65.0% O
- (c) 60.0% C, 4.4% H, and the remainder O
- 3.88** The compound XCl_4 contains 75.0% Cl by mass. What is the element X?
- 3.89** What is the molecular formula of each of the following compounds?
- (a) empirical formula CH_3O , molar mass = 62.0 g/mol
- (b) empirical formula NH_2 , molar mass = 32.0 g/mol
- 3.90** Determine the empirical and molecular formulas of each of the following substances:
- (a) Ibuprofen, a headache remedy, contains 75.69% C, 8.80% H, and 15.51% O by mass and has a molar mass of 206 g/mol.
- (b) Cadaverine, a foul-smelling substance produced by the action of bacteria on meat, contains 58.55% C, 13.81% H, and 27.40% N by mass; its molar mass is 102.2 g/mol.
- (c) Epinephrine (adrenaline), a hormone secreted into the bloodstream in times of danger or stress, contains 59.0% C, 7.1% H, 26.2% O, and 7.7% N by mass; its molar mass is about 180 u.
- 3.91** (a) The characteristic odor of pineapple is due to ethyl butyrate, a compound containing carbon, hydrogen, and oxygen. Combustion of 2.78 mg of ethyl butyrate produces 6.32 mg of CO_2 and 2.58 mg of H_2O . What is the empirical formula of the compound? (b) Nicotine, a component of tobacco, is composed of C, H, and N. A 5.250-mg sample of nicotine was combusted, producing 14.242 mg of CO_2 and 4.083 mg of H_2O . What is the empirical formula for nicotine? If nicotine has a molar mass of 160 ± 5 g/mol, what is its molecular formula?
- 3.92** Propenoic acid, $C_3H_4O_2$, is a reactive organic liquid that is used in the manufacturing of plastics, coatings, and adhesives. An unlabeled container is thought to contain this liquid. A 0.275-g sample of the liquid is combusted to produce 0.102 g of water and 0.374 g carbon dioxide. Is the unknown liquid propenoic acid? Support your reasoning with calculations.
- 3.93** Epsom salts, a strong laxative used in veterinary medicine, is a hydrate, which means that a certain number of water molecules are included in the solid structure. The formula for Epsom salts can be written as $MgSO_4 \cdot xH_2O$, where x indicates the number of moles of H_2O per mole of $MgSO_4$. When 5.061 g of this hydrate is heated to 250 °C, all the water of hydration is lost, leaving 2.472 g of $MgSO_4$. What is the value of x ?

Reaction Stoichiometry (Section 3.6)

- 3.94** The reaction between potassium superoxide, KO_2 , and CO_2 ,



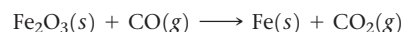
is used as a source of O_2 and absorber of CO_2 in self-contained breathing equipment used by rescue workers.

- (a) How many moles of O_2 are produced when 0.400 mol of KO_2 reacts in this fashion?
- (b) How many grams of KO_2 are needed to form 7.50 g of O_2 ?



- (c) How many grams of CO_2 are used when 7.50 g of O_2 are produced?

- 3.95** An iron ore sample contains Fe_2O_3 together with other substances. Reaction of the ore with CO produces iron metal:



- (a) Balance this equation.
- (b) Calculate the number of grams of CO that can react with 0.350 kg of Fe_2O_3 .
- (c) Calculate the number of grams of Fe and the number of grams of CO_2 formed when 0.350 kg of Fe_2O_3 reacts.
- (d) Show that your calculations in parts (b) and (c) are consistent with the law of conservation of mass.
- 3.96** Calcium hydride reacts with water to form calcium hydroxide and hydrogen gas. (a) Write a balanced chemical equation for the reaction. (b) How many grams of calcium hydride are needed to form 4.500 g of hydrogen?
- 3.97** The complete combustion of octane, C_8H_{18} , a component of gasoline, proceeds as follows:
- $$2 C_8H_{18}(l) + 25 O_2(g) \longrightarrow 16 CO_2(g) + 18 H_2O(g)$$
- (a) How many moles of O_2 are needed to burn 1.50 mol of C_8H_{18} ?
- (b) How many grams of O_2 are needed to burn 10.0 g of C_8H_{18} ?
- (c) Octane has a density of 0.692 g/mL at 20 °C. How many grams of O_2 are required to burn 15.0 gal of C_8H_{18} (the capacity of an average fuel tank)?
- (d) How many grams of CO_2 are produced when 15.0 gal of C_8H_{18} are combusted?

3.98 Detonation of nitroglycerin proceeds as follows:



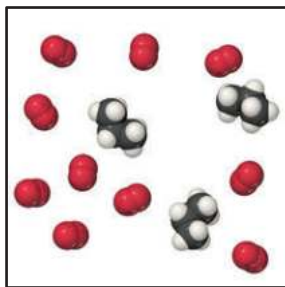
(a) If a sample containing 2.00 mL of nitroglycerin (density = 1.592 g/mL) is detonated, how many moles of gas are produced? (b) If each mole of gas occupies 55 L under the conditions of the explosion, how many liters of gas are produced? (c) How many grams of N_2 are produced in the detonation?

3.99 The combustion of one mole of liquid octane, $\text{CH}_3(\text{CH}_2)_6\text{CH}_3$, produces 5470 kJ of heat. Calculate how much heat is produced if 1.000 gallon of octane is combusted. See Exercise 3.97 for necessary information about octane.

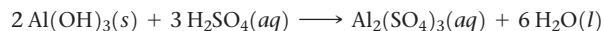
Limiting Reactants (Section 3.7)

3.100 Define the terms *theoretical yield*, *actual yield*, and *percent yield*. (b) Why is the actual yield in a reaction almost always less than the theoretical yield? (c) Can a reaction ever have 110% actual yield?

3.101 Consider the mixture of propane, C_3H_8 , and O_2 shown here. (a) Write a balanced equation for the combustion reaction that occurs between propane and oxygen. (b) Which reactant is the limiting reactant? (c) How many molecules of CO_2 , H_2O , C_3H_8 , and O_2 will be present if the reaction goes to completion?



3.102 Aluminum hydroxide reacts with sulfuric acid as follows:



Which is the limiting reactant when 0.500 mol $\text{Al}(\text{OH})_3$ and 0.500 mol H_2SO_4 are allowed to react? How many moles of the excess reactant remain after the completion of the reaction?

3.103 One of the steps in the commercial process for converting ammonia to nitric acid is the conversion of NH_3 to NO :



In a certain experiment, 2.00 g of NH_3 reacts with 2.50 g of O_2 . (a) Which is the limiting reactant? (b) How many grams of NO and H_2O form? (c) How many grams of the excess reactant remain after the limiting reactant is completely consumed? (d) Show that your calculations in parts (b) and (c) are consistent with the law of conservation of mass.

3.104 Solutions of sulfuric acid and lead(II) acetate react to form solid lead(II) sulfate and a solution of acetic acid. If 5.00 g of sulfuric acid and 5.00 g of lead(II) acetate are mixed, calculate the number of grams of sulfuric acid, lead(II) acetate, lead(II) sulfate, and acetic acid present in the mixture after the reaction is complete.

3.105 When ethane (C_2H_6) reacts with chlorine (Cl_2), the main product is $\text{C}_2\text{H}_5\text{Cl}$, but other products containing Cl, such as $\text{C}_2\text{H}_4\text{Cl}_2$, are also obtained in small quantities. The formation of these other products reduces the yield of $\text{C}_2\text{H}_5\text{Cl}$. (a) Calculate the theoretical yield of $\text{C}_2\text{H}_5\text{Cl}$ when 125 g of C_2H_6 reacts with 255 g of Cl_2 , assuming that C_2H_6 and Cl_2 react only to form $\text{C}_2\text{H}_5\text{Cl}$ and HCl . (b) Calculate the percent yield of $\text{C}_2\text{H}_5\text{Cl}$ if the reaction produces 206 g of $\text{C}_2\text{H}_5\text{Cl}$.

3.106 When hydrogen sulfide gas is bubbled into a solution of sodium hydroxide, the reaction forms sodium sulfide and water. How many grams of sodium sulfide are formed if 1.25 g of hydrogen sulfide is bubbled into a solution containing 2.00 g of sodium hydroxide, assuming that the sodium sulfide is made in 92.0% yield?

Additional Exercises

3.107 Write balanced chemical equations for (a) the complete combustion of acetone (CH_3COCH_3), a common organic solvent; (b) the decomposition of solid mercury (I) carbonate into carbon dioxide gas, mercury, and solid mercury oxide; (c) the combination reaction between sulphur dioxide gas and liquid water to produce sulfurous acid.

3.108 If 2.0 mol $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$, 2.0 mol C_4H_{10} , and 2.0 mol C_6H_6 are completely combusted in oxygen, which one produces the largest number of moles of H_2O ? Which one produces the least? Explain.

3.109 Calcium is an essential nutrient in our body. It is important for bone health. Four common calcium-containing supplements are calcium carbonate (CaCO_3), calcium citrate ($\text{Ca}_3\text{C}_{12}\text{H}_{10}\text{O}_{14}$), calcium gluconate ($\text{CaC}_{12}\text{H}_{22}\text{O}_{14}$), and calcium lactate ($\text{CaC}_6\text{H}_{10}\text{O}_6$). Rank these calcium supplements in terms of the mass percentage of calcium they contain.

3.110 (a) Ibuprofen is a common over-the-counter analgesic with the formula $\text{C}_{13}\text{H}_{18}\text{O}_2$. How many moles of $\text{C}_{13}\text{H}_{18}\text{O}_2$ are in a 500-mg tablet of ibuprofen? Assume the tablet is composed entirely of ibuprofen. (b) How many molecules of $\text{C}_{13}\text{H}_{18}\text{O}_2$ are in this tablet? (c) How many oxygen atoms are in the tablet?

3.111 Very small semiconductor crystals, composed of approximately 1000 to 10,000 atoms, are called quantum dots. Quantum dots made of the semiconductor CdSe are now being used in electronic reader and tablet displays because they emit light efficiently and in multiple colors, depending on dot size. The density of CdSe is 5.82 g/cm³.

(a) What is the mass of one 2.5-nm CdSe quantum dot?

(b) CdSe quantum dots that are 2.5 nm in diameter emit blue light upon stimulation. Assuming that the dot is a perfect sphere and that the empty space in the dot can be neglected, calculate how many Cd atoms are in one quantum dot of this size.

(c) What is the mass of one 6.5-nm CdSe quantum dot?

(d) CdSe quantum dots that are 6.5 nm in diameter emit red light upon stimulation. Assuming that the dot is a perfect sphere, calculate how many Cd atoms are in one quantum dot of this size.

(e) If you wanted to make one 6.5-nm dot from multiple 2.5-nm dots, how many 2.5-nm dots would you need, and how many CdSe formula units would be left over, if any?

- 3.112** (a) One molecule of the antibiotic penicillin G has a mass of 5.342×10^{-21} g. What is the molar mass of penicillin G? (b) Hemoglobin, the oxygen-carrying protein in red blood cells, has four iron atoms per molecule and contains 0.340% iron by mass. Calculate the molar mass of hemoglobin.
- 3.113** Cinnamaldehyde is a compound that is responsible for the characteristic aroma of cinnamon. It contains 81.79% C, 6.10% H, and the remaining is oxygen. Its molar mass is 132 g/mol. Determine its molecular formula.
- 3.114** Fructose, commonly called fruit sugar, is a monosaccharide found in many plants. It contains 40% C, 6.71% H, and the remainder O. (a) What is the empirical formula for fructose? (b) A mass spectrum of fructose shows a peak at about 180 u. What is the molecular formula of the substance?
- 3.115** Vanillin, the dominant flavoring in vanilla, contains C, H, and O. When 1.05 g of this substance is completely combusted, 2.43 g of CO_2 and 0.50 g of H_2O are produced. What is the empirical formula of vanillin?
- 3.116** An organic compound was found to contain only C, H, and Cl. When a 1.50-g sample of the compound was completely combusted in air, 3.52 g of CO_2 was formed. In a separate experiment, the chlorine in a 1.00-g sample of the compound was converted to 1.27 g of AgCl. Determine the empirical formula of the compound.
- 3.117** A compound, $\text{Na}_2\text{Cr}_2\text{O}_x$, where x is unknown, is analyzed and found to contain 39.70% Cr. What is the value of x ?
- 3.118** An element X forms an iodide (XI_3) and a chloride (XCl_3). The iodide is quantitatively converted to the chloride when it is heated in a stream of chlorine:
- $$2 \text{XI}_3 + 3 \text{Cl}_2 \longrightarrow 2 \text{XCl}_3 + 3 \text{I}_2$$
- If 0.5000 g of XI_3 is treated with chlorine, 0.2360 g of XCl_3 is obtained. (a) Calculate the atomic weight of the element X. (b) Identify the element X.
- 3.119** A method used by the U.S. Environmental Protection Agency (EPA) for determining the concentration of ozone in air is to pass the air sample through a “bubbler” containing sodium iodide, which removes the ozone according to the following equation:
- $$\text{O}_3(\text{g}) + 2 \text{NaI}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \longrightarrow \text{O}_2(\text{g}) + \text{I}_2(\text{s}) + 2 \text{NaOH}(\text{aq})$$
- (a) How many moles of sodium iodide are needed to remove 5.95×10^{-6} mol of O_3 ? (b) How many grams of sodium iodide are needed to remove 1.3 mg of O_3 ?
- 3.120** A chemical plant uses electrical energy to decompose aqueous solutions of NaCl to give Cl_2 , H_2 , and NaOH:
- $$2 \text{NaCl}(\text{aq}) + 2 \text{H}_2\text{O}(\text{l}) \longrightarrow 2 \text{NaOH}(\text{aq}) + \text{H}_2(\text{g}) + \text{Cl}_2(\text{g})$$
- If the plant produces 1.5×10^6 kg (1500 metric tons) of Cl_2 daily, estimate the quantities of H_2 and NaOH produced.
- 3.121** The fat stored in a camel's hump is a source of both energy and water. Calculate the mass of H_2O produced by the metabolism of 1.0 kg of fat, assuming the fat consists entirely of tristearin ($\text{C}_{57}\text{H}_{110}\text{O}_6$), a typical animal fat, and assuming that during metabolism, tristearin reacts with O_2 to form only CO_2 and H_2O .
- 3.122** When hydrocarbons are burned in a limited amount of air, both CO and CO_2 form. When 0.450 g of a particular hydrocarbon was burned in air, 0.467 g of CO, 0.733 g of CO_2 , and 0.450 g of H_2O were formed. (a) What is the empirical formula of the compound? (b) How many grams of O_2 were used in the reaction? (c) How many grams would have been required for complete combustion?
- 3.123** A mixture of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ reacts in a closed container to form ammonia, $\text{NH}_3(\text{g})$. The reaction ceases before either reactant has been totally consumed. At this stage 3.0 mol N_2 , 3.0 mol H_2 , and 3.0 mol NH_3 are present. How many moles of N_2 and H_2 were present originally?
- 3.124** A mixture containing KClO_3 , K_2CO_3 , KHCO_3 , and KCl was heated, producing CO_2 , O_2 , and H_2O gases according to the following equations:
- $$2 \text{KClO}_3(\text{s}) \longrightarrow 2 \text{KCl}(\text{s}) + 3 \text{O}_2(\text{g})$$
- $$2 \text{KHCO}_3(\text{s}) \longrightarrow \text{K}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{g}) + 2 \text{CO}_2(\text{g})$$
- $$\text{K}_2\text{CO}_3(\text{s}) \longrightarrow \text{K}_2\text{O}(\text{s}) + \text{CO}_2(\text{g})$$
- The KCl does not react under the conditions of the reaction. If 100.0 g of the mixture produces 1.80 g of H_2O , 13.20 g of CO_2 , and 4.00 g of O_2 , what was the composition of the original mixture? (Assume complete decomposition of the mixture.)
- 3.125** When a mixture of 10.0 g of acetylene (C_2H_2) and 10.0 g of oxygen (O_2) is ignited, the resulting combustion reaction produces CO_2 and H_2O . (a) Write the balanced chemical equation for this reaction. (b) Which is the limiting reactant? (c) How many grams of C_2H_2 , O_2 , CO_2 , and H_2O are present after the reaction is complete?

Integrative Exercises

These exercises require skills from earlier chapters as well as skills from the present chapter.

- 3.126** Boron nitride, BN, is an electrical insulator with remarkable thermal and chemical stability. Its density is 2.1 g/cm^3 . It can be made by reacting boric acid, H_3BO_3 , with ammonia. The other product of the reaction is water. (a) Write a balanced chemical equation for the synthesis of BN. (b) If you made 225 g of boric acid react with 150 g ammonia, what mass of BN could you make? (c) Which reactant, if any, would be left over, and how many moles of leftover reactant would remain? (d) One application of BN is as thin film for electrical insulation. If you take the mass of BN from part (a) and make a 0.4 mm thin film from it, what area, in cm^2 , would it cover?
- 3.127** Viridicatumtoxin B, $\text{C}_{30}\text{H}_{31}\text{NO}_{10}$, is a natural antibiotic compound. It requires a synthesis of 12 steps in the laboratory. Assuming all steps have equivalent yields of 85%, which is the final percent yield of the total synthesis?
- 3.128** Consider a sample of calcium carbonate in the form of a cube measuring 2.005 in. on each edge. If the sample has a density of 2.71 g/cm^3 , how many oxygen atoms does it contain?
- 3.129** (a) You are given a cube of silver metal that measures 1.000 cm on each edge. The density of silver is 10.5 g/cm^3 . How many atoms are in this cube? (b) Because atoms are spherical, they cannot occupy all of the space of the cube. The silver atoms pack in the solid in such a way that 74% of the volume of the solid is actually filled with the silver atoms. Calculate the volume of a single silver atom. (c) Using the volume of a silver atom and the formula for the volume of a sphere, calculate the radius in angstroms of a silver atom.
- 3.130** (a) If an automobile travels 350 km with a gas mileage of 9.0 km/L, how many kilograms of CO_2 are produced? Assume that the gasoline is composed of octane, $\text{C}_8\text{H}_{18}(\text{l})$, whose density is 0.692 g/mL . (b) Repeat the calculation for a truck that has a gas mileage of 2 km/L.
- 3.131** Section 2.9 introduced the idea of structural isomerism, with 1-propanol and 2-propanol as examples. Determine which of these properties would distinguish these two substances: (a) boiling point, (b) combustion analysis results, (c) molecular weight, (d) density at a given temperature and pressure. You can check on the properties of these two compounds in

Wolfram Alpha (<http://www.wolframalpha.com/>) or the *CRC Handbook of Chemistry and Physics*.

- 3.132** NO_x is a generic term for the nitrogen oxides, NO and NO_2 . NO_x gases are air pollutants that react to form smog and acid rain. In order to reduce NO_x emission from vehicle, catalytic converters are installed in car exhausts to decompose NO and NO_2 respectively into N_2 and O_2 . (a) Write the balanced chemical equations for the decomposition of NO and NO_2 respectively. (b) If the car produces 100 g NO_x a day, with equal mole ratio of NO and NO_2 , how many grams of NO and NO_2 are produced respectively?

- 3.133** Hydrogen cyanide, HCN , is a poisonous gas. The lethal dose is approximately 300 mg HCN per kilogram of air when inhaled. (a) Calculate the amount of HCN that gives the lethal dose in a small laboratory room measuring $3.5 \times 4.5 \times 2.5$ m. The density of air at 26°C is 0.00118 g/cm^3 . (b) If the HCN is formed by reaction of NaCN with an acid such as H_2SO_4 , what mass of NaCN gives the lethal dose in the room?

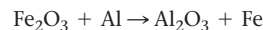


(c) HCN forms when synthetic fibers containing Orlon® or Acrilan® burn. Acrilan® has an empirical formula of CH_2CHCN , so HCN is 50.9% of the formula by mass. A rug measures 3.5×4.5 m and contains 850 g of Acrilan® fibers per square yard of carpet. If the rug burns, will a lethal dose of HCN be generated in the room? Assume that the yield of HCN from the fibers is 20% and that the carpet is 50% consumed.

- 3.134** The source of oxygen that drives the internal combustion engine in an automobile is air. Air is a mixture of gases, principally N_2 (~79%) and O_2 (~20%). In the cylinder of an automobile engine, nitrogen can react with oxygen to produce nitric oxide gas, NO . As NO is emitted from the tailpipe of the car, it can react with more oxygen to produce nitrogen dioxide gas. (a) Write balanced chemical equations for both reactions. (b) Both nitric oxide and nitrogen dioxide are pollutants that can lead to acid rain and global warming; collectively, they are called “ NO_x ” gases. In 2009, the United States emitted an estimated 19 million tons of nitrogen dioxide into the atmosphere. How many grams of nitrogen dioxide is this? (c) The production of NO_x gases is an unwanted side reaction of the main engine combustion process that turns octane, C_8H_{18} , into CO_2 and water. If 85% of the oxygen in an engine is used to combust octane and the

remainder used to produce nitrogen dioxide, calculate how many grams of nitrogen dioxide would be produced during the combustion of 500 g of octane.

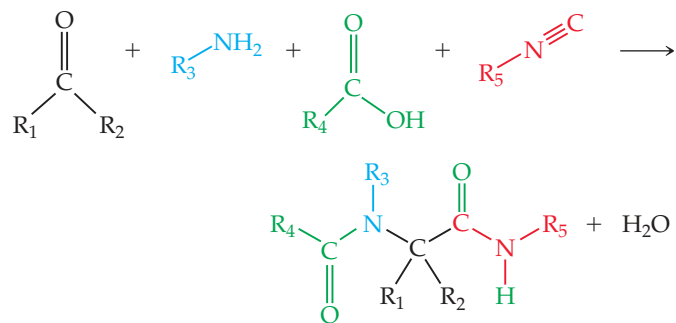
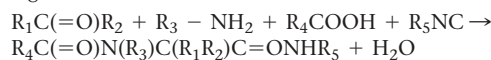
- 3.135** The thermite reaction,



produces so much heat that the Fe product melts. This reaction is used industrially to weld metal parts under water, where a torch cannot be employed. It is also a favorite chemical demonstration in the lecture hall (on a small scale).

- (a) Balance the chemical equation for the thermite reaction, and include the proper states of matter. (b) Calculate how many grams of aluminum are needed to completely react with 500.0 g of Fe_2O_3 in this reaction. (c) This reaction produces 852 kJ of heat per mole of Fe_2O_3 reacted. How many grams of Fe_2O_3 are needed to produce 1.00×10^4 kJ of heat? (d) If you performed the reverse reaction—aluminum oxide plus iron makes iron oxide plus aluminum—would that reaction have heat as a reactant or a product?

- 3.136** One of the most bizarre reactions in chemistry is called the Ugi reaction:



- (a) Write out the balanced chemical equation for the Ugi reaction, for the case where $\text{R} = \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ (this is called the *hexyl* group) for all compounds. (b) What mass of the “hexyl Ugi product” would you form if 435.0 mg of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ was the limiting reactant?

Design an Experiment

You will learn later in this book that sulfur is capable of forming two common oxides, SO_2 and SO_3 . One question that we might ask is whether the direct reaction between sulfur and oxygen leads to the formation of SO_2 , SO_3 , or a mixture of the two. This question has practical significance because SO_3 can go on to react with water to form sulfuric acid, H_2SO_4 , which is produced industrially on a very large scale. Consider also that the answer to this question may depend on the relative amount of each element that is present and the temperature at which the reaction is carried out. For example, on the one hand, carbon and oxygen normally react to form CO_2 , but when not enough oxygen is present, CO can form. On the other hand, under normal reaction conditions H_2 and O_2 react to form water, H_2O (rather than hydrogen peroxide H_2O_2), regardless of the starting ratio of hydrogen to oxygen.

Suppose you are given a bottle of sulfur, which is a yellow solid, a cylinder of O_2 , a transparent reaction vessel that can be evacuated and sealed so that only sulfur, oxygen, and the product(s) of the reaction between the two are present, an analytical balance so

that you can determine the masses of the reactants and/or products, and a furnace that can be used to heat the reaction vessel to 200°C where the two elements react. (a) If you start with 0.10 mol of sulfur in the reaction vessel how many moles of oxygen would need to be added to form SO_2 , assuming SO_2 forms exclusively? (b) How many moles of oxygen would be needed to form SO_3 , assuming SO_3 forms exclusively? (c) Given the available equipment, how would you determine you had added the correct number of moles of each reactant to the reaction vessel? (d) What observation or experimental technique would you use to determine the identity of the reaction product(s)? Could differences in the physical properties of SO_2 and SO_3 be used to help identify the product(s)? Have any instruments been described in Chapters 1–3 that would allow you to identify the product(s)? (e) What experiments would you conduct to determine if the product(s) of this reaction (either SO_2 or SO_3 or a mixture of the two) can be controlled by varying the ratio of sulfur and oxygen that are added to the reaction vessel? What ratio(s) of S to O_2 would you test to answer this question?